Microarray-Based Biomarker Discovery for Steroid-Induced Osteonecrosis of the Femoral Head (SONFH) using an Automated Machine Learning Pipeline

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Course / Module: Omics Logic Artificial Intelligence and Machine Learning

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https://github.com/dvrk-dvys/Omics_Capstone

Abstract

The disease Steroid-Induced Osteonecrosis of the Femoral Head (SONFH) as a glucocorticoid-induced avascular necrosis condition is routinely diagnosed late leading to progressive bone degeneration and femoral head collapse. The unmet clinical need of early molecular detection leads to the diagnosis of the disease being irreversible and detrimental to the quality of life of its sufferers. There are many candidates for reliable blood-based biomarkers that are being currently worked on but ultimately the findings remain limited or inconsistent in current literature. This study applied 4 independent supervised ML frameworks and an Automated Retrieval Augmented Generation(RAG) pipeline to identify and qualify academic evidence in public research and academic databases to validate the findings of the ML models. This GSE123568 (17, 3) microarray dataset (40 samples: 30 SONFH, 10 control; Affymetrix GPL15207) was asses and the 4 independent pipelines: a manual Weka baseline and an automated Python AutoOmics pipeline, each evaluated under multivariate hybrid-score feature selection and univariate ANN-based ranking. Which produced the candidate gene shortlist to be processed by the RAG app (GPT-4o-mini) and assign tier-based biological evidence. All pipelines achieved strong classification performance, with Auto-Weka reaching 97.5% accuracy (AUC 0.950) and the Python multivariate pipeline achieving near-perfect AUC in tuned models. GYPA, TSTA3, and PIP5K1B were consistently identified across both multivariate pipelines, while NLRP1 was independently recovered by both univariate pipelines; notably, GYPA and TSTA3 are supported by Tier 1 direct biological evidence. In contrast, PIK3CD represents the only gene directly aligning with the original source study, consistent with prior evidence implicating PI3K/AKT signaling in SONFH pathogenesis.The pipeline was able to successfully reassess an older dataset (2018) to identify genes of interest that only recent research papers have been able to prove are integral to SOFNH through its flexible Supervised ML followed by a robust automated self reasoning retrieval system.

1. Introduction

1.1 SONFH Background

Steroid-Induced Osteonecrosis of the Femoral Head (SONFH) is a severe debilitating joint and bone condition that is characterized by reduced blood supply to the hip socket (acetabulum) leading to bone death (osteonecrosis) and even a full joint collapse. The disease is known to be caused by commonly prescribed glucosteroid medication used to treat autoimmune-related and inflammatory diseases such as prednisone for Allergy Induced Asthma and others (1). The condition also can be triggered by reduced blood supply (ischemia), altered lipid metabolism and oxidative stress caused by prolonged alcohol abuse (2). Pharmacological interventions and experimental blood serum transfusions have been promising but without early detection the condition is usually irreversible and requires surgical intervention such as a full joint replacement. Recent research efforts have been able to find promising potential serum biomarkers for early detection but more work is still needed. This paper serves as an exploratory research analysis to showcase robust machine learning and automation techniques for identifying and qualifying biomarkers for SONFH detection. I have chosen to base this report off of the GSE123568 (3) data set where of the 40 surveyed patients: 30 had developed SONFH after prolonged use of steroids while 10 control patients developed no adverse symptoms after steroid usage.

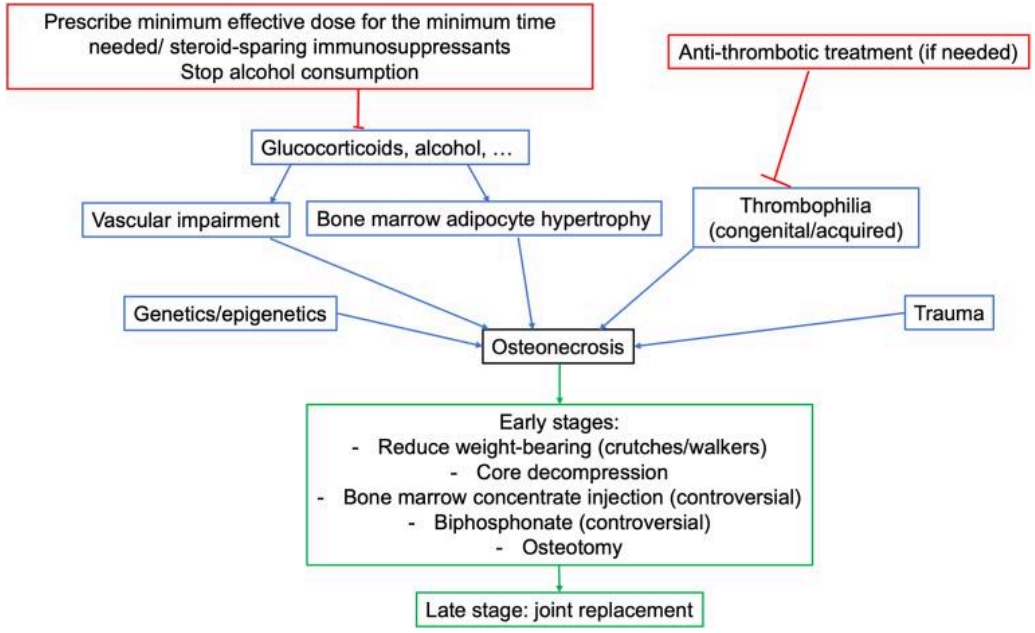


Figure 1. Overview of Steroid-Induced Osteonecrosis of the Femoral Head (SONFH): disease mechanism, clinical staging, and the progression from microvascular disruption to femoral head collapse. The progressive and often irreversible nature of the condition underscores the clinical need for early molecular biomarker detection. (4)

1.2 Rationale for Transcriptomic / Microarray Data

Unfortunately because of recent and ongoing research in the field, the most robust datasets and results have not been made fully public. So I have had difficulty finding a GEO dataset that has enough patients and raw data to actually be viable for training a machine learning model to a statistically valuable result. So I decided

to rely on the dataset that has the most patients with the trade off of switching to using microarray data (GSE123568) rather than RNA-seq data as mentioned in the rubric because of insufficient patient-level samples, and several were single-cell datasets unsuitable for bulk-level supervised classification. Because of this we cannot discover novel isoforms, but my ML approach is suitable enough toward classification goals and uses a custom automated ML pipeline to rank and research genes of interest from the microarray of 49,293 gene probes these researchers used (3).

1.3 Rationale for Both Weka and Automated Machine Learning and Objectives

Due to the extreme dimensionality imbalance in the dataset (49,293 probes vs 40 samples), there is a high risk of noise, overfitting, and the potential loss of biologically relevant genes during preprocessing and feature selection. While Weka provides a strong and accessible framework, I also wanted to evaluate how a more automated, python-based approach would perform on the same data. After running a suite of Weka models, including Auto-Weka, I developed an automated Omics ML pipeline to mirror the same core steps, enabling direct comparison and combination of results. Both pipelines apply two feature selection strategies, a multivariate hybrid score (fold change plus t-statistic) and a univariate ANN wrapper, with the goal of maximising recovery of biologically relevant candidate genes supported by downstream Agentic RAG evidence.The objective of this study is to analyse GSE123568 using both Weka and an automated ML pipeline to identify differentially expressed genes, determine the best-performing model, rank probes by their contribution to SONFH classification across both feature selection strategies, generate and compare biomarker shortlists, and interpret their biological relevance using an LLM-powered Agentic RAG system grounded in existing literature.

2. Materials and Methods

As outlined in the Introduction, the GSE123568 dataset was selected due to its high probe dimensionality and sufficient sample size (30 SONFH, 10 controls), providing both biological relevance and enough class representation for supervised learning. The dataset was obtained from the NCBI GEO repository and generated using Affymetrix microarray technology, producing an initial matrix of 49,293 probes across 40 samples. After variance filtering (IQR < 0.5 $\log_2$ units), 11,687 probes were retained. The data were already $\log_2$-transformed and normalised, requiring no further transformation. Then preprocessing scripts were used to prepare the data for modelling. These steps directly emulate the logic of the R scripts provided in the course, ensuring consistency across both the Weka and automated Python pipelines. This includes parsing the GEO matrix into a samples × probes format with class labels, followed by variance-based filtering to remove low-information probes. A stricter threshold was used for the Weka pipeline, while a more permissive filter was applied in the automated pipeline to allow a broader feature search.

2.2 Feature Selection Strategies

Univariate (ANN Wrapper Approach): A univariate ANN wrapper approach was implemented, as described in the course and provided in the *simple ANN wrapper and filter.R* file, to evaluate each probe independently. For each probe, a small neural network model was trained using only that single feature as input, following repeated stratified Monte Carlo cross-validation splits (typically 70% train / 30% test). Model performance was evaluated on held-out test data, and probes were ranked based on their median test AUC across all runs. This produces a ranking of the form:

$$Score_{probe} = median(AUC_{test}^{(i)}),\ i = 1,\ldots, N$$

Where ***N*** is the number of repeated Monte Carlo splits.

The ANN-based wrapper was retained as the primary method due to its stable and interpretable performance. The top 100 probes were selected after this univariate feature selection for both the manual Weka-based and AutoOmics univariate pipelines. This approach isolates genes with strong independent predictive signals, but does not capture interactions between genes.An optional reranking step was also tested to break ties among probes with similarly high ANN performance. In this variant, probes were reranked using:

$$Univariate\ (ReRanker)\ Score = 0.7 \times z(AUC) + 0.3 \times z(|FC|)$$

These weights allow genes with both high AUC and large fold change to rise to the top of the ranking.

Multivariate (Hybrid Scoring Approach): In contrast to the univariate ANN wrapper, the *feature_select.py* script uses a hybrid ranking score that evaluates all probes simultaneously using differential expression statistics:

$$Hybrid\ Score = Z(|FC|) + Z(|t|)$$

Where:

- ***FC*** (fold change) is the difference in mean $\log_2$ expression between SONFH and control groups ($\boldsymbol{FC} = \mu_{SONFH} - \mu_{control}$);
- ***t*** is Welch's t-statistic comparing probe expression between groups;
- ***Z***(·) denotes z-score normalisation across all probes.

This approach prioritises probes that are both strongly differentially expressed (effect size) and statistically significant across all samples. The top 100 probes were selected for the Weka-based classification, while the AutoOmics Python pipeline was tested with both top 100 and top 500 probes. My intuition behind expanding the feature set from 100 to 500 was that for the multivariate results a wider pool of lower-ranked features would reduce the dominance of higher-ranked signals, allowing additional potentially significant but weaker-signalled gene patterns to contribute to the model performance and make it into the final biomarker shortlist (described in Section 2.5). Output files were generated in .arff format for direct compatibility with Weka.

2.3 Weka Pipeline - Manual Baseline

After the feature selection strategies in Section 2.2 were run as standalone scripts, the formatted .arff input file is ready for Weka. I then uploaded and ran a full suite of classification algorithms which were evaluated under a consistent cross-validation framework, and model performance was compared using standard metrics (Accuracy, AUC ROC, Weighted F1-score, Kappa) to identify the optimal model. The best model, Random Forest, was chosen after a comparative analysis (see Sections 3.3–3.4), then the J48 Decision Tree is used to split nodes so any probes used at the decision boundaries are added to the biomarker shortlist, which serves as the input into the Agentic RAG enrichment job.

2.4 AutoOmics Pipeline and LLM Evidence Integration

The AutoOmics pipeline extends the core methodology into a fully automated and reproducible system, operating in two modes (multivariate and univariate) as seen above, under a repeated cross-validation scheme. Balanced accuracy is used as the primary evaluation metric to account for class imbalance. Across both modes, models are evaluated iteratively, and the best-performing configuration is used to derive feature importance and generate biomarker shortlists for downstream analysis. These shortlists are passed into an Agentic Retrieval Augmented Generation (RAG) stage, designed to function as a constrained research assistant rather than a decision-maker. The system does not confirm biomarkers; instead, it assembles structured evidence to support human-in-the-loop evaluation. The RAG agent operates as an iterative retrieval–reasoning loop. For each candidate gene, it first resolves canonical identity and aliases, then performs a sequence of targeted

retrievals across biological and biomedical sources. Retrieved information is ranked by semantic similarity to the query to ensure that only the most relevant evidence is retained within the context window.This process repeats for multiple iterations, allowing the agent to accumulate and refine evidence across steps before producing a final structured output(JSON). With tier-labelled evidence profiles for each gene, preserving both the strength and type of supporting information. This enables consistent comparison across pipeline outputs and provides a transparent basis for downstream biological interpretation and validation.

2.6 Evaluation Framework - Predictive Performance and Biological Validity

The pipeline was evaluated at multiple levels because no single metric fully captures biomarker quality. Different stages were designed to optimise different objectives: statistical feature selection was used to detect strong expression signals, classification metrics were used to assess predictive performance, and tier-based evidence review was used to assess biological validity.

Predictive performance metrics: AUC (area under the ROC curve), balanced accuracy, weighted F1-score. Balanced accuracy was prioritised in the AutoOmics pipeline due to class imbalance (30:10).

> **Metric caveat:** Weka reports *standard* accuracy; the Python pipeline reports *balanced* accuracy. These are not directly comparable on a 30:10 imbalanced dataset. A model predicting the majority class only would achieve 75% standard accuracy but only 50% balanced accuracy. This distinction is addressed explicitly in the Discussion (Section 4.5).

Biological validity - tier-based evidence framework:

- **Tier 1** : human biomarker evidence directly tied to the active disease context, matching the study modality (e.g. peripheral blood microarray) when possible.
- **Tier 2** : human disease-specific mechanistic, network, or association evidence.
- **Tier 3** : human evidence in disease-adjacent biology. "Disease-adjacent" is contextual — inferred from the active disease name, GEO dataset context, tissue/source, and retrieved evidence.
- **Tier 4** : animal, cell-line, speculative, or otherwise weakly grounded support only.

Relational evidence types:

- *direct* : retrieved sources explicitly link this gene to the active disease;
- *indirect* : no direct disease paper, but biologically relevant pathway or mechanism support exists;
- *inferred* : reasoning from general gene biology with weak disease-specific grounding.

Interpretive principles: high model importance does not always equal high biological importance; total gene count ≠ pipeline quality; correlated genes may share the same biological signal, causing predictive models to prioritise only one representative feature (see Section 4.3). This evaluation framework enables comparison not only of predictive performance, but also of biological validity distinguishing pipelines that identify statistically predictive features from those that also recover literature-supported biomarkers, and assessing the quality of that literature, not just supposition of predictive features.

2.7 Pipeline Overview

See the full diagram pipeline here

Figure 2. Figure 2 summarises the full AutoOmics workflow, from shared preprocessing through parallel feature-selection and classification branches to LLM-based evidence integration.

3. Results

3.1 Dataset Summary

The GSE123568 dataset consists of 40 patient-level microarray samples, 30 SONFH cases and 10 non-SONFH controls, giving a 3:1 class imbalance that motivated the use of balanced accuracy in the Python pipeline. The raw matrix contained 49,293 probes, of which 45,465 were retained after variance filtering (IQR < 0.2 $\log_2$ units). PCA of the filtered matrix (Figure 3) showed that PC1 and PC2 explained 88.0% and 2.4% of the variance respectively (90.4% total), indicating a strong biological signal separating the two classes. As this study uses microarray rather than RNA-seq data, expression values reflect probe hybridisation intensity rather than direct transcript counts; despite limitations such as probe cross-hybridisation and reduced dynamic range, the resulting matrix remained suitable for downstream machine learning analysis.

3.2 Exploratory Data Analysis

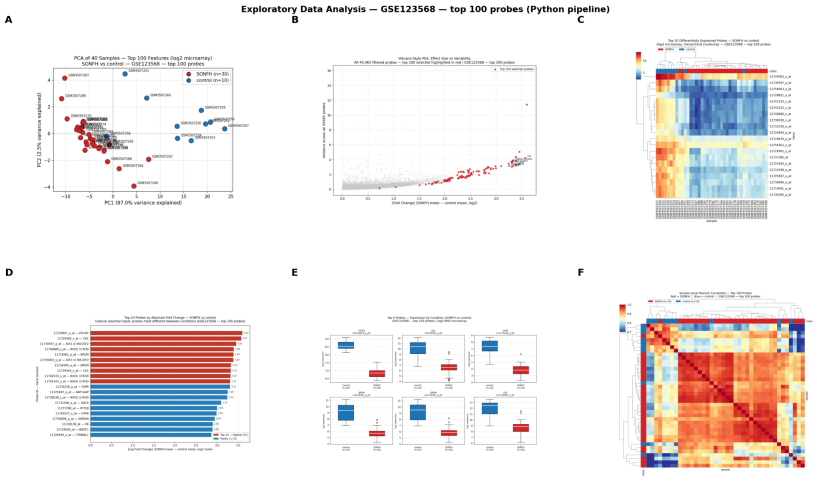


Figure 3. Multi-panel EDA of GSE123568 (shared preprocessing step, applied before pipeline branching). (A) PCA: PC1 = 88.0%, PC2 = 2.4%. (B) Volcano plot of differential expression (SONFH vs control). (C) Heatmap, top 20 probes by variance. (D) Fold-change bar chart, top 20 probes. (E) Boxplots, top 6 probes by class. (F) Sample-level Pearson correlation heatmap. n=40 (30 SONFH, 10 control).

I would consider the EDA suite that I ran to be very helpful and serve good explainable evidence that supports the use of the Supervised ML approaches I chose to employ. The key plots to notice here are:

- **Plot A:** The selected features show a strong signal separating the two groups, with near clean separation along PC1 (87% variance explained).Overall, the clear separation suggests that the dataset contains a learnable structure, supporting the use of supervised machine learning for classification.
- **Plot C:** Heatmap (clustered expression) : This plot shows the two feature classes with samples clearly clustering together when all points are displayed together. The diseased people have similar expression profiles in the same probes and the opposite is also true for the control. Clear boundary & good signal for the medical interpretation.
- **Plot F:** Pearson Correlation Matrix : This plot aims to see if the samples are internally consistent. And again through the separation of two feature classes we see within-class similarities that reinforce the findings of the PCA.

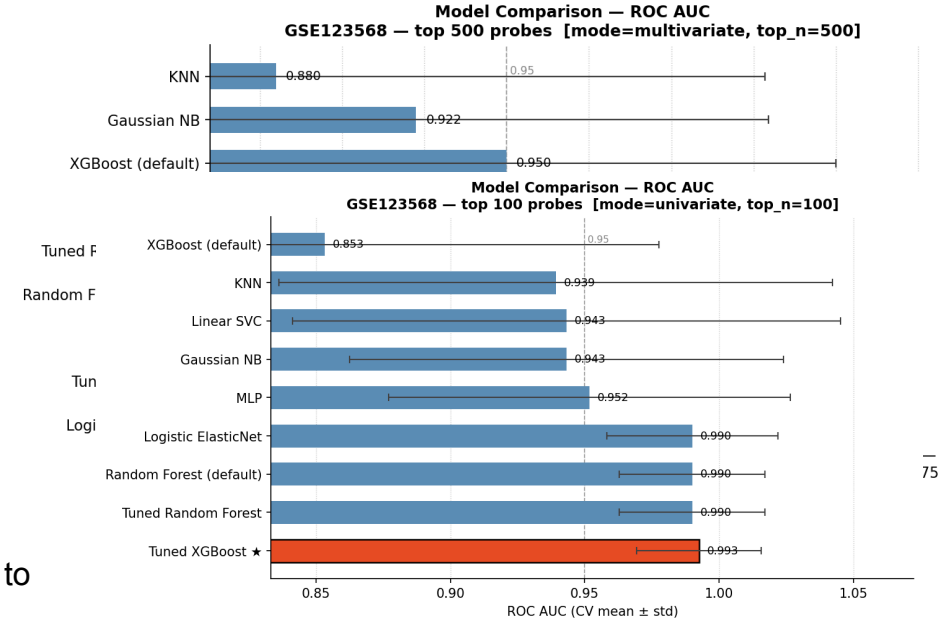
The other plots (B:Volcano Plot, D:Top 20 fold change and E:Top 6 Probes) were just exploratory walks for myself to see which probe:gene pairs had the highest representation or severity in the expression profiles to see initially what to look out for. In these you'll notice BPGM repeated many times through all plots rising to the top. We will speak more about it in section 3.9.

3.3 Weka Multivariate - Model Performance

The three best performing models were the Multilayer Perceptron (ANN), Random Forest (200 trees) & AutoWeka using a PART rule learner with CfsSubsetEval attribute selection. Due to overfitting on other models I went ahead with the Random Forest attribute importance ranking score to rank all the probes. It is worth noting though that all models exceeded 90% standard accuracy on this dataset.The univariate model performance had nearly the exact same efficacy leading me to choosing the exact same model to maintain pipeline parity. So I chose to choose between the different configurations I was testing. See **Table 2** (in the appendix)

3.5 Python Multivariate — Model Performance (Top-500 Hybrid Score Ranked Features: Best Run)

Figure 6. Python multivariate classifier performance of the best run (top-500 hybrid-score features). ROC AUC (mean ± std) using RepeatedStratifiedKFold (5×10) on top-500 features. The Hyperopt-tuned XGBoost model is highlighted.



With this suite of models and the use of hyperopt i was able to achieve impressive results but I was faced with a major problem of interpretability; Is the accuracy of this model pointing to an optimization of the models own performance, or to the learning of the most biologically important probe:gene pairs to the disease classification. Therein lies the major issue, some runs of the Hyperopt Xgboost would lead to very high scores but when I would manually research the top genes that were found none of them were known to be established biomarkers of osteonecrosis of the femoral head (ONFH) or any of its modalities. So after a lot of research and consulting I chose to stick with the Random Forest because I understand that it reflects average impurity reduction across many independent trees, providing a more stable and aggregated view of feature contribution. Since feature selection precedes model training, all models operate on the same pre-filtered feature space. However, XGBoost's optimising for classification performance, may concentrate importance on whichever features most efficiently reduce training error, which on a small dataset risks overfitting to a few dominant probes, whereas Random Forest distributes importance more evenly across independent trees *making it better suited* for identifying a *wider set of robust biomarker candidates* rather than optimising just for classification alone. As a result, the sequential optimisation in XGBoost risks over-emphasising this initial distribution, potentially prioritising features that improve classification performance without corresponding biological relevance.

3.6 Python Univariate — Model Performance

Here, despite strong performance from multiple models, again Random Forest was chosen to maintain consistency with the biomarker prioritisation stage, where it is used to compute feature importance. This ensures alignment between feature ranking and downstream biological interpretation, rather than selecting a model based solely on predictive performance. While the Hyperopt-tuned XGBoost model achieved the highest predictive performance, model choice did not influence feature selection, as probes were pre-ranked prior to training.

3.7 Full Cross-Pipeline Biomarker Audit

Both Weka and Python pipelines independently generated biomarker shortlists under both multivariate and univariate feature selection strategies. After which LLM-based tier filtering was applied (*direct OR indirect, Tier 1–3*) and 43 unique genes were identified across all four pipelines. This has been collated and presented in **Table 6** (in the appendix) which presents the complete audit; Each cell reports evidence tier, feature selection ranking score (i.e., the pipeline-specific score assigned to the probe within its originating feature selection method), and relation type (d = direct, i = indirect); KnownA corresponds to curated SONFH biomarkers that I collected through my own research and formatted to be used in the report. This list also served as the basis for the creation of the *Tier metric* used in the Agentic RAG system. KnownB corresponds to *Jia et al. (2023)* (3) stage-specific biomarkers which is the paper that was produced by the same researchers and lab that did the wet lab biomarker assay data base that this report is based off of; The Notes column provides high-level functional annotations (*e.g., erythroid biology, immune signaling, apoptosis*) to contextualize the biological relevance of each candidate. |FC| represents absolute $\log_2$ fold change (SONFH vs control). KnownA corresponds to curated SONFH biomarkers that I collected through my own research and formatted to be used in the report. Green rows denote Tier 1 candidates; yellow rows indicate genes identified by ≥2 pipelines (cross-pipeline convergence). Eleven of these genes which are highlighted in green were confirmed in curated SONFH known-gene literature (*BPGM, CISD2, GYPA, HEMGN, NLRP1, PIP5K1B, TMCC2, TSTA3, BNIP3L, PIK3CD, RUNX2*) and two of them represent novel candidates not previously catalogued in either curated known gene set (*PIP5K1B, TSTA3*). **Table 7** (in the appendix) summarises per-pipeline counts, and from this you can see that theMultivariate pipelines not only recovered more Tier 1 biomarkers but also exhibited higher average fold-change magnitude, indicating stronger biological signal capture but maybe too strong mapping to strong signals and crowding out more subtle signatures. More on this in the discussion.

3.8 Selected Candidate Gene Profiles

NLRP1 — *Univariate overlap* ***· Tier 1(Univariate: Top-100, Top-500 & Weka)***

Now we switch to the top univariate exclusive gene, NLRP1, with **Tier 1 direct evidence** and **|FC| ≈ 1.27**. This gene represents pyroptosis and inflammatory signaling, suggesting that the univariate models recovered a complementary inflammatory signature that the multivariate models did not prioritize. Maybe because it doesn't have many other complementary genes to create a signature across multiple probes. A key paper states that "NLRP1 had the highest diagnostic value" and is one of the strongest "potential pyroptosis-related biomarkers." (12) The finding of this gene shows that different pipelines recover different but biologically meaningful aspects of SONFH.

BPGM — *Multivariate overlap* ***· Tier 1 (Multivariate: Top-100, Top-500 & Weka)***

It is very appropriate to start the list of highlight candidate genes with one of the most commonly known and most researched genes connected to SONFH and especially because it was one of the top genes to be surfaced during the EDA phase (see section 3.1). This gene was seen across all Multivariate model runs with a **high feature-selection score** with one of the largest absolute Fold Change values (***|FC| ≈ 3.39***). It follows that the AgenticRAG evidence layer found many **Tier-1** quality citations which all point to this genes lower expression in SONFH individuals as a gene that points to the symptoms of the disease such as femoral head ischemia, hypoxia and altered circulation. In healthy physiology, BPGM is an erythrocyte metabolic gene linked to oxygen handling. The most informative citation found by AutoOmics states that "BPGM … was downregulated in SONFH patients" and "might be used as diagnostic biomarkers." It also states that "GYPA, TMCC2, and BPGM were identified as potential biomarkers" in peripheral serum (9); all of which were also found by the system, although they are all similar erythrocyte metabolic genes. (See **Table 6** in the appendix) It is my opinion though that this and other erythrocyte genes may show irregularities too late in the development stages of SONFH and may only point to symptomatic effects of the disease and may be a function of the mechanical traumas caused by a socket collapse having already begun, therefore making it not so much of a suitable gene for early detection of the disease.

CISD2 — *Multivariate overlap* ***· Tier 1 (Multivariate: Top-100, Top-500 & Weka)***

This gene was also found in all Multivariate model runs but with a much weaker **feature-selection score** in Top-100 & Top-500 but maintaining a good signal in the Weka Multivariate model. There was still a **strong fold change (|FC| ≈ 2.41)**, making it one of the most consistent non-erythroid multivariate findings. CISAD2 points toward ferroptosis, mitochondria dysfunction, and oxidative stress, all of which fit Steroid Induced-ONFH and even Alcohol Induced-pathophysiology. The top citation found by the AutoOmics app supports the idea that "...CISD2, NCOA4, and GCLC (are) potential diagnostic biomarkers associated with immune cell infiltration in

SONFH, offering new insights for future research and clinical applications.'. The paper further states that CISD2 is "significantly lower in SONFH bone tissue" and that it is "significantly involved in inflammatory pathways.". Lastly, the paper mentions that "Mitochondrial dysfunction is central to" SONFH pathogenesis, "releasing excess ROS and apoptosis-inducing factors" (10). This, to me, is a great candidate for early detection and upstream mechanism of SONFH. This gene reinforces the claim that the pipeline can recover biologically coherent candidates beyond obvious or previously well-documented genes.

BNIP3L — Singular Multivariate · Tier 2 (Weka Multivariate ONLY)

BNIP3L was selected to be featured, despite its Tier 2 ranking and indirect evidence because it was the highest scoring gene to be found by only the Weka Multivariate pipeline. This gene plays an important role in apoptosis, mitophagy and hypoxia mechanism that fits the known pathology of SONFH. My intuition is that BNIP3L helps represent the mitochondrial stress side of the disease and complements CISD2. A found paper had BNIP3L among a suite of other genes to diagnose SONFH and states that when it is downregulated it leads to or is a signal of **"cellular response to hypoxia"** and **"Mitochondrion".** This is a good candidate for more research that is an alternative to erythrocyte detection. (14)

RUNX2 — Singular Univariate · Tier 1 (Python Top 100 univariate ONLY)

RUNX2 is a key bone regulator though affecting transcription factors for osteoblast differentiation. It is another gene that was solely flagged by the Python Top-100 Univariate approach **with Tier 1 direct evidence and |FC| ≈ 1.22**, and is one of the few representing the osteogenic axis of SONFH pathology. This paper states that in a diseased state (sic) "expressions of Runx2 directly were decreased", and that this provides strong mechanistic evidence linking SONFH progression to impaired osteogenesis and angiogenesis (16). Interestingly enough another paper AutoOmics found an atorvastatin study which states that though "upregulation of osteogenic genes (Runx2)", "ATO can be utilized in the prevention of SONFH, thereby holding significant implications for the prevention and treatment of SONFH in clinical settings.". This is a common anticholesterol medication that doctors should know could be used to also help prevent SONFH. (15)

PIK3CD — Singular Univarite · Tier 2 (Python Top 100 univariate ONLY) 🎉

The PIK3CD gene is of particular interest to me because it is the one and only gene that made it into the top 100 or top 500 rankings of any model type that is a gene that is directly written about in the same research paper that was written by the same research team that produced the dataset that this entire AutoOmics system was trained on (GSE123568 - 3) . It was ranked as a Tier 2 gene with indirect related mechanistic evidence that was only rank high enough to be cataloged by the univariate Top-100 pipeline. I investigated this gene's signal thoroughly and I found that this gene didn't have particularly high signal in the micro array (like the symptomatic erythrogenic genes) *because* of the fact that it may be a true upstream causal gene for SONFH. Which the keystone paper of this project states clearly;

“the relative expression levels of PRKCA, **PIK3CD** and CD4 mRNA were all **dramatically up-regulated in the NONFH early-stage patients** compared with the non-NONFH control group (NC group) and other stage groups”. This finding was experimentally validated through (RT-qPCR) and I am glad that my AutoOmics pipeline was also able to find this gene as one of interest even if it wasn't rated highly. The paper actually employed a very different preprocessing approach that is specific to the research institution that the researchers are from. They utilized many private gene databases to construct a unified gene interaction network and employed ‘Cytoscape v3.8.2 software to calculate the degree, betweenness centrality and closeness centrality’ within a single combined gene interaction network. (3) Completely different approach, but i was able to get an overlap. This is the strongest signal of the strength of the system. With more time and effort one could expand farther than top 500 into the full reranked list of 49,293 probes.

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4. Discussion

4.1 Principal Findings

This ML MicroArray analysis applied tailormade preprocessing algorithms before training to optimize the output of the models achieving strong classification performance. Both Weka & Python candidate models were trained to +97% accuracy and +0.950 AUC, but the Random Forest Classification approach was chosen for all 4 pipes. The results were 43 total literature confirmed genes of interest and full LLM generated reasoning and research documentation produced in an easily ingestiable .JSON format by my AutoOmics App. Many key SONFH pathology modalities (erythroid Mitochondrial, immune-inflammatory ect) were recovered and with pipeline results combined paints a complete biological picture of SONFH and could help researchers and doctors find key genes that may be worth more study to help with early detection and preventative measures. I would like to assert the possibility that some of the other genes that did not have strong evidence found for them but were highly ranked may actually be candidates for completely new research.

4.2 Biological Interpretation — Downstream vs Upstream Signals

The biological signal recovered in this study appears to operate across multiple layers rather than pointing to one single disease pathway. First, the strongest and loudest cluster consisted of erythroid-associated genes such as **GYPA, BPGM, HEMGN, and TSTA3**. These genes are highly correlated and likely reflect downstream physiological disruption caused by SONFH, particularly hypoxia, vascular compromise, and altered bone marrow conditions. This interpretation is consistent with known disease mechanisms involving impaired blood supply and ischemia in the femoral head (1). In other words, these may be some of the clearest symptomatic signals in the dataset, but not necessarily the original drivers of the disease. A second layer is represented by **CISD2**, which point toward mitochondrial

stress, ferroptosis, oxidative damage, and lipid dysregulation. These findings align well with established models of steroid-induced metabolic injury and oxidative stress contributing to osteonecrosis (2, 10, 11). Finally, **NLRP1, RUNX2, and PIK3CD** may sit closer to upstream disease mechanisms. These genes suggest inflammatory activation, impaired osteogenesis, and early-stage immune signaling, all of which have been independently associated with SONFH progression and early-stage pathology (3, 12, 15). Taken together, this suggests that multivariate pipelines tend to recover the loud downstream wreckage, while univariate methods are better at surfacing smaller, more mechanistically interesting upstream signals that warrant further investigation. These findings are in line with the care paper produced by the producers of this data set and suggest that my models were able to find some distinctions between early mid and late biomarker signals. (3)

4.3 ML Feature Importance vs Biological Importance

A key distinction in this study is that machine learning feature importance reflects predictive utility, not necessarily biological causality. A gene ranked highly by a Random Forest model simply means it was useful for separating SONFH from control samples and not that it is a causal driver of the disease. Th HyperOpt and XGBoost model results show this clearly. The models were trained to near perfection but the LLM outputs were close to meaningless except for some strong generic biomarker signals being found. This becomes especially important in high-dimensional omics data, where many genes are highly correlated. For example, erythroid genes such as **BPGM, GYPA** both reflect similar underlying biological processes. The model only needs one of these to make accurate splits, and will typically select whichever feature performs best locally in this case often **TSTA3**. This is an overgeneralizing that often leads to the model choosing to prioritize a signal that isn't actually biologically relevant. This is a well-known effect in high-dimensional feature spaces. High predictive accuracy achieved with a small feature subset does not imply that only those features are biologically relevant but implies that the disease signal is redundantly represented across many correlated genes. This was adjusted for by sticking to the interpretable Random forest and tailoring each feature selection calculations to the actual question being asked: ‘Which probe could be a key signal for the detection of SONFH’ and then ‘which gene:probe pairs are actually upstream detectors?’

4.4 Multivariate vs Univariate Feature Selection

The two feature selection strategies used in this study address fundamentally different questions. The multivariate hybrid score asks: ‘*which genes work best together?’* In contrast, the univariate ANN wrapper asks: ‘*which genes are individually predictive?’* These are not equivalent, and the results reflect that distinction. The multivariate approach consistently recovered a coordinated erythroid cluster (**GYPA, BPGM, TSTA3, HEMGN**), as these genes exhibit strong fold change and co-vary across samples, forming a dominant shared signal. In contrast, the univariate approach identified **PIK3CD**, which has strong individual early detection power but does not carry strong blood serum expression which means its signal doesn’t align with the common and well documented erythroid cluster. Because of this, it is

not prioritised by the hybrid score, but emerges clearly when evaluated independently. **GYPA** appears in both approaches, suggesting it is both individually predictive and part of the broader coordinated signal, making it one of the most robust candidates. Overall, the two methods recover complementary biological layers, and combining them produces a more complete and balanced biomarker set.

4.5 Comparison with Published Literature

When comparing these findings to the source study for GSE123568 (3), the fact that my system didn't find the same genes of interest is not contradictory, but methodologically expected. Their pipeline prioritised gene interaction networks and hub centrality using Cytoscape software, meaning the genes were selected based on connectivity within a protein:protein interaction framework. In contrast, my AutoOmics pipeline prioritized statistical signal through fold change and machine learning feature importance. Our approaches are fundamentally asking different questions: one identifies network regulators, the other identifies expression-dominant signals. This distinction explains why genes such as **PIK3CD** appear in both analyses but are ranked differently. In my study, PIK3CD was present but ranked low due to modest fold change, suggesting it represents an early-stage or regulatory signal rather than a dominant expression marker. This aligns with its role described in the original paper, confirming presence without dominance.

Beyond the source study, several high-confidence genes identified here are also supported by independent literature. Overall, the results show strong consistency with multiple strands of SONFH literature while also highlighting method-dependent differences in gene prioritisation. Rather than conflicting, these approaches should be viewed as complementary, each capturing different layers of disease biology ranging from network-level regulators to dominant expression signals. And all of this was done in an automated fashion using LLM powered AutoOmics app.

4.6 Limitations

This study has several important limitations that should be acknowledged. First, the dataset is relatively small (n = 40) with a 30:10 class imbalance, which limits statistical power and may inflate standard accuracy metrics. Second, all findings are based on a single microarray dataset with no external validation cohort, meaning the results should be considered exploratory rather than definitive. Third, microarray platforms inherently constrain analysis through predefined probe sets, potential cross-hybridisation (_x_at probes), and reduced dynamic range compared to RNA-seq. Additionally, some signals such as EIF1AY (Y-chromosome gene) suggest possible confounding variables (e.g. sex imbalance) that were not explicitly controlled for. There are also methodological differences between pipelines (Top-100 vs Top-500 feature sets, 10-fold vs repeated CV) which make comparisons approximate rather than perfectly equivalent. Finally, all LLM-assisted scientific validation was performed through an and could be prone to cost explosions due to the looping nature of the LLM reasoning. No functional or experimental validation was

performed, so all candidates remain expression-level associations and the LLM research done only serves to help inform the direction other scientists may follow when looking into SONFH.

4.7 Future Directions

There are several clear next steps that could strengthen and expand this work. First, validation of key candidates such as **GYPA, NLRP1, and PIK3CD** in a larger, more balanced, and independent cohort would be critical to confirm reproducibility. Second, functional validation studies should be conducted to test the biological role of these genes, particularly upstream candidates such as NLRP1 (inflammatory signaling) and RUNX2 (osteogenesis) in relevant cell or tissue models. A particularly important extension of this work would be to systematically analyze genes that rank highly in model outputs but receive lower evidence tiers due to limited existing literature. For example, **PIK3CD** appears as a low fold-change, early-stage signal that is not strongly expressed but is known from prior studies to be mechanistically relevant to SONFH. This suggests that some biologically important genes may be underrepresented in purely expression-driven rankings. Similarly, highly ranked Tier 3-4 genes with sparse literature may represent underexplored or emerging biological signals rather than noise. Future work could prioritise these genes for targeted wet-lab validation, helping to identify novel biomarkers that are currently overlooked due to publication bias.
In addition, incorporating prior biological knowledge into preprocessing such as differentiating known upstream causal genes from downstream symptomatic signals could help guide feature selection toward more mechanistically informative candidates. This could be achieved through hybrid scoring approaches that combine statistical signals with curated biological prior knowledge. Finally, pathway-level integration and multi-omics analysis (e.g. proteomics, metabolomics) could further connect downstream erythroid signals with upstream immune and metabolic mechanisms, producing a more complete systems-level understanding of SONFH.

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5. Conclusion

This study demonstrates that combining multiple machine learning pipelines with structured LLM-based biological validation can produce a more complete and interpretable view of complex disease biology. My AutoOmics system was designed as an automated and reproducible framework, integrating four parallel pipelines (Weka and Python, each with multivariate and univariate feature selection) into a cohesive automated workflow. This required extensive effort in preprocessing, standardization, feature selection design, model tuning, and output structuring, ultimately enabling consistent comparison across fundamentally different analytical approaches. Across these pipelines, 14 highlighted candidate genes were identified, with consistent multi-pipeline support for **GYPA, BPGM, TSTA3**, **PIK3CD** and **NLRP1**, and Tier 1 evidence confirming several as literature-supported biomarkers of SONFH. Rather than converging on a single pathway,

the results reveal a layered biological structure: strong downstream erythroid signals reflecting vascular and hypoxic disruption, intermediate metabolic and mitochondrial stress signals, and more subtle upstream candidates linked to inflammation, immune regulation, and osteogenesis. By explicitly separating predictive performance from biological credibility through tier-based evaluation, this work provides a more interpretable framework for high-dimensional omics analysis. The automated nature of AutoOmics ensures that this process is not only reproducible but also can be configured for other GEO datasets and disease contexts. Ultimately, this approach not only reproduces known SONFH biology but also surfaces less-studied candidates, bridging the gap between statistical signal detection and biologically actionable insight.

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Appendix

Table 2. Weka multivariate: 10-fold stratified CV, top-100 hybrid-score features (n=40). Green = best performers. Standard accuracy reported (Not balanced; please see metric caveat, Section 2.6). The Univariate results are nearly the same. This is an abbreviated table full results can be found in the git repo.

Model	Accuracy (%)	F1 (Wt.)	ROC AUC	Notes
Auto-Weka	**97.5**	**0.975**	**0.950**	**Best overall; PART + CfsSubsetEval; 1 error**
J48	95.0	0.948	0.820	Single split on PIP5K1B (≥4.915), used to produce the final shortlist
Random Forest	**92.5**	**0.924**	**0.950**	**Tied highest AUC; top features: RHD, RHCE**
SMO (SVM)	90.0	0.896	0.833	Lowest AUC overall
IBk (k=1)	90.0	0.896	0.833	4 misclassifications

Table 4. Python multivariate w/ RepeatedStratifiedKFold (5×10), top-500 hybrid-score features (n=40). Green = best AUC. (T) = Hyperopt-tuned.

Model	AUC Mean	Bal. Acc. Mean	F1 (Wt.) STD	Accuracy Mean	Notes
Random Forest (T)	***0.9075***	***0.8483***	***0.1096***	***0.9075***	**Hyperopt somehow reduced accuracy here**
Random Forest	0.9817	0.8383	0.1132	0.9025	**Chosen Model , for all Hybrid Score Runs**
MLP	0.98	0.9333	0.0639	0.95	No native feature importance metric for biomarker analysis
XGBoost (T)	0.9992	0.95	0.0631	0.97	**Best AUC**
Logistic (ElasticNet)	1.0	0.9333	0.0638	0.955	**Overfitting**
XGBoost	0.95	0.95	0.0564	0.975	

Table 5. Python univariate ANN — RepeatedStratifiedKFold (5×10), top-100 ANN-ranked features (n=40).

Model	AUC Mean	Bal. Acc. Mean	F1 (Wt.) STD	Accuracy Mean	Notes
Random Forest (T)	***0.99***	***0.9067***	***0.077***	***0.935***	**Hyperopt helped the accuracy mean here**
Random Forest	**0.99**	**0.8983**	**0.077**	**0.9325**	**Chosen Model , for all Hybrid Score Runs**
XGBoost (T)	0.8533	0.895	0.083	0.9275	**Hyperopt hurt this models performance in Univariate Mode**
Logistic (ElasticNet)	0.99	0.8833	0.084	0.925	
XGBoost	0.95	0.8533	0.0804	0.905	

Table 6. Full cross-pipeline biomarker audit of 43 candidate genes across six pipeline configurations (M100, M500, U100, UR100, WM, WU) after LLM-based evidence filtering (Tier 1–3; direct/indirect). Each cell reports evidence tier, feature selection ranking score (pipeline-specific), and relation type (d/i); |FC| = absolute $\log_2$ fold change (SONFH vs control); # of Pipelines = cross-run frequency; KnownA/B = literature matches; Notes = functional annotation.

Gene	M100	M500	U100	UR100	WM	WU	FC	# of Pipelines	KnownA	KnownB	Notes
BPGM	T1(0.68)d	T1(0.66)d	—	—	T1(0.65)d	—	3.388	3	✓	—	Erythroid Metabolism
CISD2	T1(0.45)d	T1(0.37)d	—	—	T1(0.71)d	—	2.414	3	—	—	Ferroptosis / Mitochondrial Biology
GYPA	T1(0.54)d	T1(0.51)d	—	—	T1(0.73)d	—	3.306	3	✓	—	Erythroid Surface
HEMGN	T1(0.46)d	T1(0.48)d	—	—	T1(0.61)d	—	2.944	3	✓	—	Erythroid TF
NLRP1	—	—	T1(1.00)d	T1(1.26)d	—	T1(0.64)d	1.266	3	✓	—	Pyroptosis
PIP5K1B	T1(0.56)d	T1(0.47)d	—	—	T1(0.65)d	—	2.02	3	—	—	Lipid Signaling
TMCC2	T2(0.53)d	T2(0.49)d	—	—	T2(0.64)d	—	2.76	3	✓	—	Erythroid Metabolism
TSTA3	T1(0.74)d	T1(0.77)d	—	—	T1(0.71)d	—	1.933	3	—	—	Immune Biomarker
ABCG2	T2(0.32)i	—	—	—	T2(0.00)i	—	2.33	2	—	—	
IGF2	T3(0.53)i	T3(0.49)i	—	—	—	—	3.452	2	—	—	
NOTCH2	—	—	T2(1.00)i	—	—	T2(0.57)i	1.127	2	—	—	
P2RY13	—	—	—	T1(1.29)d	—	T1(0.60)d	1.289	2	—	—	
STOM	T2(0.39)d	—	—	—	T2(0.43)d	—	1.627	2	—	—	
TCF3	T2(0.43)i	—	—	—	T2(0.46)i	—	1.508	2	—	—	
ABCG1	—	—	—	T2(1.41)i	—	—	1.413	1	—	—	
ARG2	—	T1(0.35)d	—	—	—	—	1.789	1	—	—	

ARNTL	—	—	T2(1.00)i	—	—	—	0.944	1	—	—	
BNIP3L	—	—	—	—	T2(0.70)i	—	1.833	1	✓	—	Apoptosis Mitophagy
CBL	—	—	T2(1.00)d	—	—	—	1.015	1	—	—	
CCR2	—	—	—	T2(1.35)i	—	—	1.349	1	—	—	
CPPED1	—	—	T3(1.00)i	—	—	—	1.398	1	—	—	
CXCR2	—	—	—	T2(1.67)i	—	—	1.677	1	—	—	
DNAJC6	—	T2(0.56)i	—	—	—	—	2.78	1	—	—	
DPEP2	—	—	T2(1.00)i	—	—	—	1.502	1	—	—	
ELOVL6	—	T1(0.41)d	—	—	—	—	1.492	1	—	—	
F2RL1	—	—	—	T2(1.44)i	—	—	1.445	1	—	—	
FCHSD2	—	—	T3(1.00)i	—	—	—	0.482	1	—	—	
GRB2	—	—	—	—	—	T2(0.69)i	0.818	1	—	—	
IL6R	—	—	—	T2(1.36)i	—	—	1.369	1	—	—	
IMPDH1	—	—	T3(1.00)i	—	—	—	1.032	1	—	—	
LCP1	—	—	T1(1.00)d	—	—	—	0.931	1	—	—	
MAP4K5	—	T1(0.40)d	—	—	—	—	1.32	1	—	—	
NOTCH1	—	—	—	—	—	T2(0.88)d	1.062	1	—	—	
PIK3CD	—	—	T2(1.00)i	—	—	—	1.105	1	—	✓	Early-stage Biomarker! 🎉
PTAFR	—	—	—	T2(1.31)d	—	—	1.31	1	—	—	
RHAG	—	—	—	—	T2(0.72)i	—	1.694	1	—	—	

RUNX2	—	—	—	—	—	T1(0.56)d	1.225	1	✓	—	Osteoblast TF
SEMA4D	—	—	—	—	—	T2(0.59)i	0.964	1	—	—	
SETD1B	—	—	T2(1.00)d	—	—	—	0.719	1	—	—	
SIRPA	—	—	—	—	—	T2(0.72)i	1.336	1	—	—	
UBE2H	—	—	—	—	T2(0.56)i	—	2.258	1	—	—	
VCAN	—	—	—	T2(1.63)i	—	—	1.635	1	—	—	
WLS	—	—	—	T2(1.41)i	—	—	1.755	1	—	—	

Table 7. Per-pipeline summary — genes passing LLM direct-evidence filter (Tier 1 + Tier 2, direct AND indirect). Note: Weka multi and Python multi (top-500) are tied on total direct-evidence genes and Tier 1 count — supporting the choice of top-500 features as the best Python run.

RUN	M100	M500	U100	UR100	WM	WU	ALL
T1_direct	6	9	2	2	6	3	13
T1_indirect	0	0	0	0	0	0	0
T2_direct	2	1	2	1	2	1	6
T2_indirect	2	1	4	7	5	4	20
T3_direct	0	0	0	0	0	0	0
T3_indirect	1	1	3	0	0	0	4
TOTAL GENES	11	12	11	10	13	8	43
AVG SCORE	0.5109	0.4975	1	1.4122	0.5823	0.6562	0.879
TIER 1 COUNT	4	4	1	1	5	2	7
TIER 2 COUNT	0	0	1	0	0	0	1
OVERALL SCORE	0.5305	0.4919	1				0.8826